PROMPT LIFE · V0.27.15

Visual Aid Audit

Rendered mobile-width screenshots and review notes for all 39 Journey visual aids. Screenshots were captured at 390px wide from the review route.

39	Total visual aids
4	Image assets
33	Coded SVG
2	Coded HTML/CSS
6	P1 future repairs
7	P2 watch/revise

Top 10 Future Repair Candidates

1. **Borrowed Flames** (collective-intelligence-lantern, P1): Important provenance/labor idea is still abstract; future visual should make data, labor, and extraction relationships more concrete.
2. **Benefit Tiers** (benefits-tool-garden, P1): Benefit tiers need sharper connection to realistic use plus human review.
3. **Cost Ledger** (costs-invisible-factory, P1): Cost categories are important and can become crowded; future pass should make costs easier to scan.
4. **Accountability Flow** (human-centered-ai-garden, P1): Accountability flow should be made more operational and less metaphor-dependent.
5. **Better-AI Control Panel** (responsible-ai-forked-path, P1): Governance/control-panel idea should be clearer for institutional decision makers.
6. **Full Chain Map** (synthesis-map-compass-lantern, P1): Full-chain synthesis is the densest visual; needs simplification before badge-facing review.
7. **Overfitting vs Generalization** (overfitting-generalization, P2): Set-aside validation is explained, but future polish should make the held-back examples visually unmistakable at 320px.
8. **Fine-Tuning Path** (before-morning-finetuning-path, P2): Generated image is usable, but a future coded callout should contrast durable adaptation with prompt, RAG, and sampling more explicitly.

9. **Text to Tokens** (tokenization, P2): Recent token-chip repair is mostly readable; keep watching the final token/punctuation example at 320px.
10. **Feature Vector** (vectors, P2): Abstract feature dimensions are correct but could use a less dense future visual for nontechnical learners.

Card-by-Card Inventory

1 Prompt to Prediction

before-morning-llm-cloud · generated PNG or image asset · P3 · Keep

BEFORE-MORNING-LLM-CLOUD

Prompt to Prediction

Lesson: What Is an LLM?

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-llm-cloud.png

Type: generated-image

Prompt to Prediction

Score, choose, append, repeat

Current context enters the model, learned weights shape next-token probabilities, one token is generated, and the response grows token by token.

1. **Context enters** The current prompt and response-so-far become the visible input for this run.
2. **Weights shape probabilities** Learned weights, built earlier during training, shape next-token scores.
3. **One token is generated** The model chooses one response token from the probability cloud

Objective	Show current context flowing through learned weights into one generated token.
Recommendation	Generated asset works well as the opening mechanism visual; keep text in HTML captions.
Sample labels	Image-based or no text labels captured

2 AI Family Tree

ai-family-tree · coded SVG · P3 · Keep

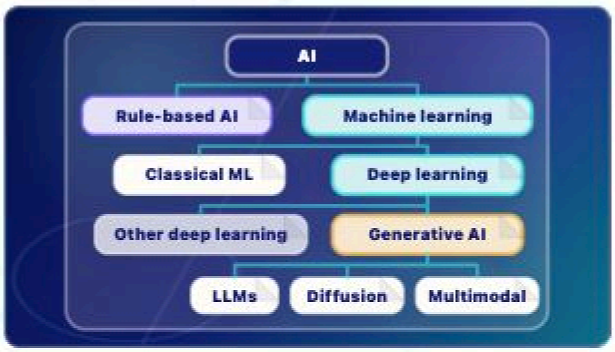
AI-FAMILY-TREE

AI Family Tree

Lesson: Where LLMs Fit

Pattern: aiTopology

Variant: origami-object



AI Family Tree

Where LLMs fit

A clean taxonomy tree shows AI as the broad field, machine learning as one branch inside AI, and LLMs as one branch inside generative AI.

1. **AI** AI is the broad field.
2. **Machine learning** Machine learning systems learn patterns from data.
3. **Deep learning** Deep learning is a neural-network branch of machine learning.
4. **Generative AI** Generative AI creates new media, such as text, images, audio, code, or video.
5. **LLMs** LLMs generate language/code; diffusion and multimodal systems are neighboring generative AI branches.

> **Key takeaway:**
> An LLM is one kind of generative AI inside the broader AI family.

Objective	Give learners a simple topology of AI categories before the history card introduces rules-first and learned-pattern traditions.
Recommendation	Readable topology visual; tap/reveal behavior supports the category map.
Sample labels	AI | Rule-based AI | Machine learning | Classical ML | Deep learning | Other deep learning | Generative AI | LLMs | Diffusion | Multimodal

3 Rules and Learned Patterns

traditions · coded SVG · P3 · Keep

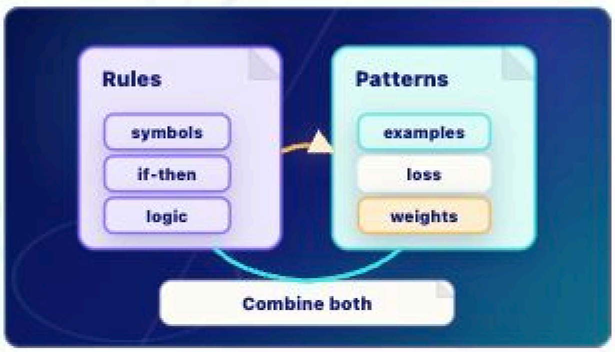
TRADITIONS

Rules and Learned Patterns

Lesson: Rationalists vs Empiricists

Pattern: traditions

Variant: paper-diagram



Rules and Learned Patterns

Two traditions, one modern toolkit

Rules-first AI uses symbols and if-then logic; deep learning learns from examples by predicting, measuring loss, and updating weights.

1. **Rules** Symbolic systems use explicit if/then logic and symbols.
2. **Examples** Loss is the training signal: it measures prediction error so weights can be adjusted.
3. **Bridge** Modern systems often combine learned models with rules, retrieval, tools, filters, and policies.

> **Key takeaway:**
> Modern AI often blends learned patterns with hand-built rules and tools.

LEARNING OBJECTIVE

Contrast explicit rules with learned patterns

Objective	Contrast explicit rules with learned patterns without turning the diagram into a poster.
Recommendation	Rules versus learned-patterns visual is clear; Loss has glossary support.
Sample labels	Rules | symbols | if-then | logic | Patterns | examples | loss | weights | Combine both

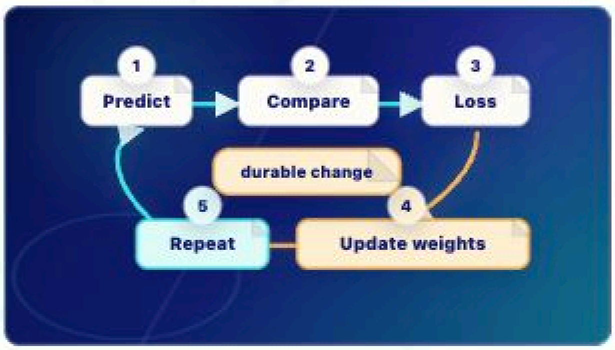
4 Training Loop

training-loop · coded SVG · P3 · Keep

TRAINING-LOOP

Training Loop

Lesson: Training | Pattern: training | Variant: paper-diagram



Training Loop

Durable change happens at weight update

Predict, compare, loss, update weights, repeat. Training changes weights.

1. **Predict** The model predicts a target.
2. **Compare** The prediction is compared with the target.
3. **Loss** Loss measures error.
4. **Update weights** Weight updates are the durable-change step.
5. **Repeat** The loop repeats many times.

> **Key takeaway:**
> Training changes weights; ordinary inference does not.

LEARNING OBJECTIVE

Show the sequence of training and make the durable weight-update step visually distinct.

ACCESSIBLE DESCRIPTION

A five-step loop moves from Predict to

Objective	Show the sequence of training and make the durable weight-update step visually distinct.
Recommendation	Training loop clearly separates examples, loss, and weight update.
Sample labels	Predict | Compare | Loss | Update weights | Repeat | 1 | 2 | 3 | 4 | 5 | durable change

5 Broad Pretraining

before-morning-pretraining-landscape · generated PNG or image asset · P3 · Keep

BEFORE-MORNING-PRETRAINING-LANDSCAPE

Broad Pretraining

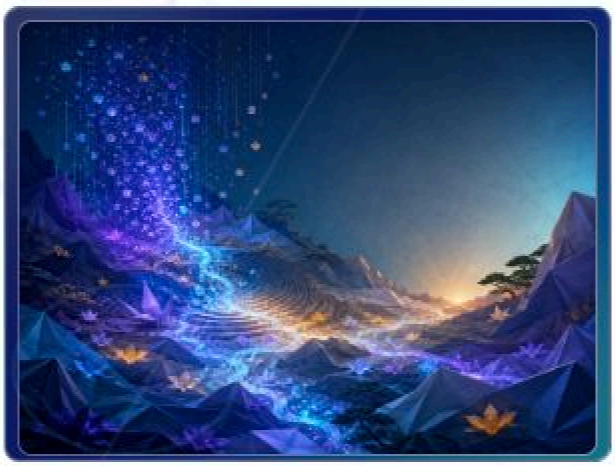
Lesson: Pretraining

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-pretraining-landscape.png

Type: generated-image



Broad Pretraining

Scale, not perfect recall

Many examples flow through the training loop. Repeated updates shape weights into broad patterns the model can use later. This does not make the model a perfect memory of its sources.

1. **Many examples flow** Pretraining repeats the training loop across enormous collections of examples.
2. **Updates shape weights** Prediction errors drive repeated weight updates during training.
3. **Broad patterns form** The model learns statistical patterns that can help later prompts.

Objective	Show broad data exposure shaping durable model weights before ordinary use.
Recommendation	Generated landscape communicates scale without replacing accessible HTML copy.
Sample labels	Image-based or no text labels captured

6 Overfitting vs Generalization

overfitting-generalization · coded SVG · P2 · Revise later

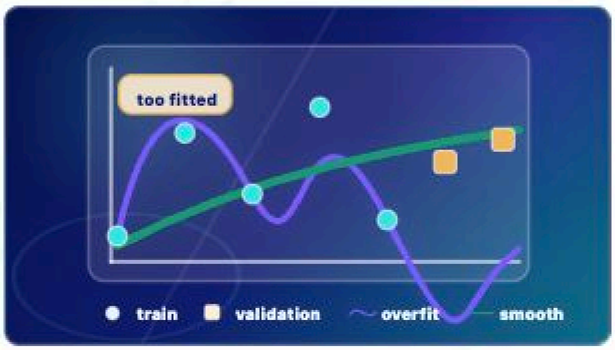
OVERFITTING-GENERALIZATION

Overfitting vs Generalization

Lesson: Overfitting and Generalization

Pattern: overfitting

Variant: zen-garden-map



Overfitting vs Generalization

Memorizing training examples is not the same as learning patterns that transfer to set-aside validation examples.

1. **Point 1** Training examples are old dots the model fit during training.
2. **Point 2** Point 2: Validation examples are set aside - kept out of training - to test transfer.
3. **Point 3** The overfit curve traces old dots too tightly.
4. **Point 4** The generalizing curve is smoother and reaches new examples.

LEARNING OBJECTIVE

Memorizing training examples is not the same as learning patterns that transfer to set-aside validation examples.

ACCESSIBLE DESCRIPTION

Memorizing training examples is not the same

Objective	Memorizing training examples is not the same as learning patterns that transfer to set-aside validation examples.
Recommendation	Set-aside validation is explained, but future polish should make the held-back examples visually unmistakable at 320px.
Sample labels	too fitted | train | validation | overfit | smooth

7 Fine-Tuning Path

before-morning-finetuning-path · generated PNG or image asset · P2 · Revise later

BEFORE-MORNING-FINETUNING-PATH

Fine-Tuning Path

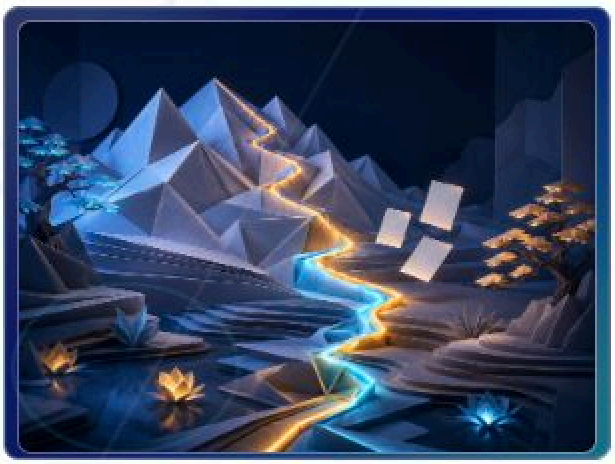
Lesson: Fine-Tuning

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-finetuning-path.png

Type: generated-image



Fine-Tuning Path

Durable adaptation after pretraining

Fine-tuning starts from a pretrained base, adds targeted examples or preference data, and shapes future responses more durably than one prompt.

1. **Pretrained base** Fine-tuning begins with a model already shaped by broad pretraining.
2. **Targeted data** Examples, domain data, or preference data guide the additional training step.
3. **Future responses** The training changes how the model is likely to behave later.
4. **More than a prompt** Fine-tuning is durable adaptation, unlike one temporary prompt or retrieved document.

Objective	Show targeted training carving a more specific path through a pretrained model landscape.
Recommendation	Generated image is usable, but a future coded callout should contrast durable adaptation with prompt, RAG, and sampling more explicitly.
Sample labels	Image-based or no text labels captured

8

Alignment Landscape

before-morning-alignment-garden · generated PNG or image asset · P3 · Keep

BEFORE-MORNING-ALIGNMENT-GARDEN

Alignment Landscape

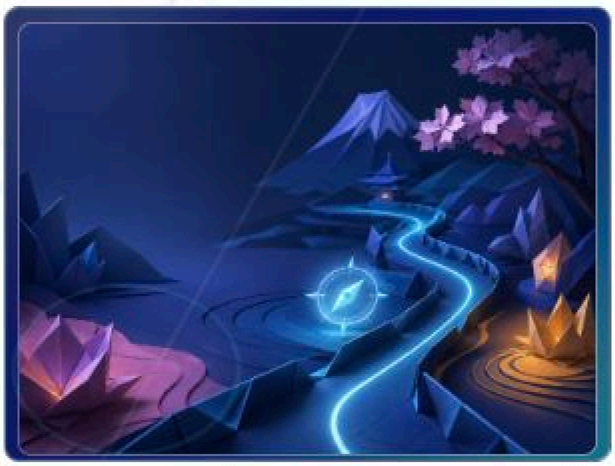
Lesson: Alignment

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-alignment-garden.png

Type: generated-image



Alignment Landscape

Shaping behavior, not conscience

Alignment encourages preferred behavior, uses guardrails to reduce risky paths, and relies on feedback and policies without giving the model a conscience.

1. **Preferred behavior** Training and system design can encourage more useful response patterns.
2. **Guardrails** Runtime safeguards can reduce risky paths during use.
3. **Feedback and policies** Human feedback, evaluations, policies, and filters can shape behavior.
4. **No conscience** Alignment does not give

Objective	Show alignment as guided behavior shaping, not moral understanding.
Recommendation	Generated visual supports the alignment metaphor; captions keep limits explicit.
Sample labels	Image-based or no text labels captured

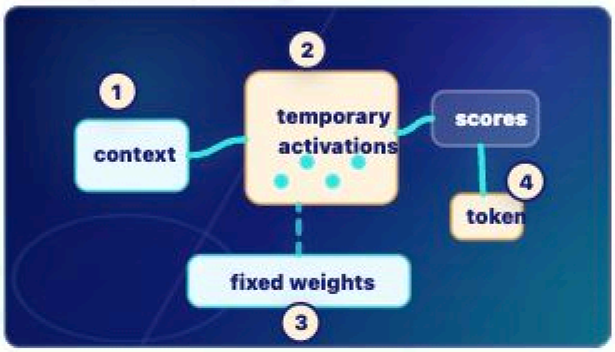
9 Forward Pass

inference-pass · coded SVG · P3 · Fixed this pass

INFERENCE-PASS

Forward Pass

Lesson: Inference | Pattern: inference | Variant: neon-flow



Forward Pass

Fixed weights, temporary activations

Inference uses fixed weights to create temporary activations, hidden states, and next-token scores.

1. **Context** The current prompt, retrieved text, conversation, and response-so-far enter the run.
2. **Fixed weights** The learned weights are used but not normally updated.
3. **Temporary activations** The run creates temporary internal states that lead to next-token scores.
4. **Next token** Decoding chooses one response token, then the loop can repeat.

> **Key takeaway:**
> Inference is a forward pass, not durable training.

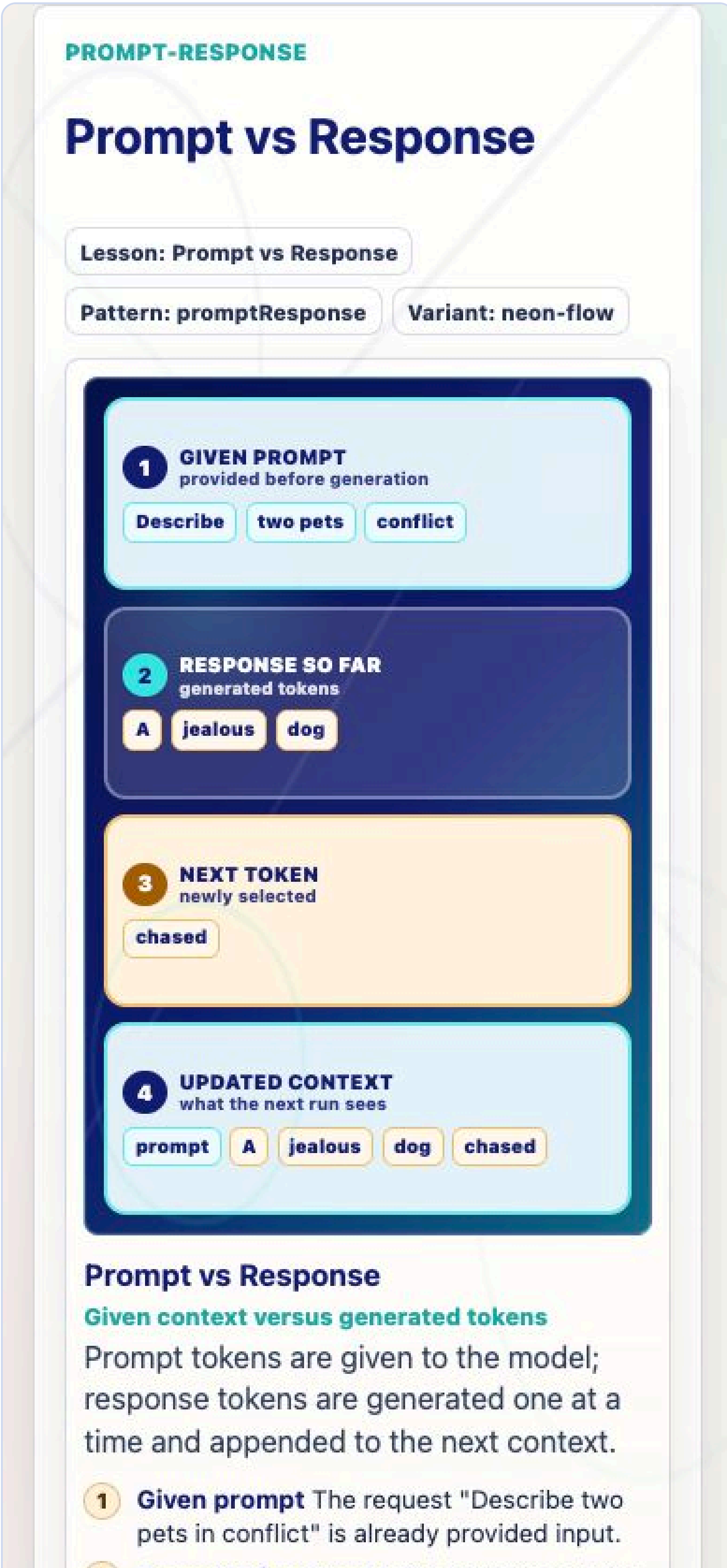
LEARNING OBJECTIVE

Make temporary inference state central while keeping durable weights visibly fixed.

Objective	Make temporary inference state central while keeping durable weights visibly fixed.
Recommendation	Tiny output label changed from the example word to token so it reads as generated-token output, not a mysterious component.
Sample labels	context | temporary | activations | scores | token | fixed weights | 1 | 2 | 3 | 4

10 Prompt vs Response

prompt-response · coded HTML/CSS · P3 · Keep



Objective	Separate the complete user prompt from response-so-far, next token, and the current context for the next run.
Recommendation	Recent stacked-panel repair is crisp and separates given prompt, response-so-far, next token, and updated context.
Sample labels	Describe | two pets | conflict | A | jealous | dog | chased | prompt | A | jealous | dog | chased

11 Text to Tokens

tokenization · coded HTML/CSS · P2 · Watch

TOKENIZATION

Text to Tokens

Lesson: Tokenization | Pattern: token | Variant: paper-diagram

Text to Tokens

Uneven chunks, punctuation included

Text is split into model-readable chunks before token IDs and embedding lookup.

1. **Simplified split** A | jealous | dog | chased | a | startled | cat | across | the | kitchen | floor | .
2. **Uneven chunks** Some tokenizers may split words or punctuation, such as start | led or floor | .
3. **Teaching note** This app uses a simplified

Objective	Show that tokens can be words, word pieces, punctuation, or other chunks.
Recommendation	Recent token-chip repair is mostly readable; keep watching the final token/punctuation example at 320px.
Sample labels	A | jealous | dog | chased | a | startled | cat | across | the | kitchen | floor | . | startled | start | led | floor. | floor | .

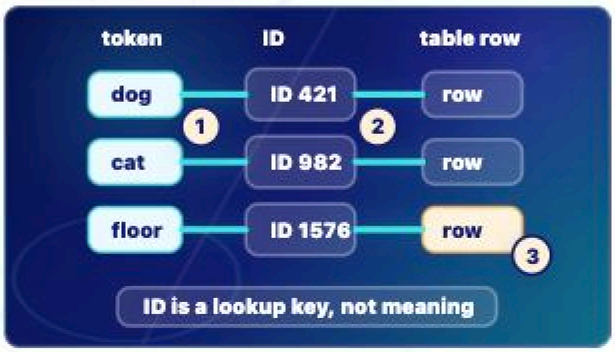
12 Token IDs

token-ids · coded SVG · P3 · Keep

TOKEN-IDS

Token IDs

Lesson: Token IDs | Pattern: ids | Variant: paper-diagram



Token IDs

Lookup keys, not meaning

Each token gets a lookup number that points to an embedding-table row.

1. **Token** dog, cat, and floor are text chunks.
2. **ID** 421, 982, and 1576 are lookup keys.
3. **Embedding row** The ID points to the learned starting vector row.
4. **Limit** The number itself is not the meaning.

Key takeaway:
Token IDs point; they do not understand.

LEARNING OBJECTIVE

Separate the token string, its numeric ID, and the embedding row the ID selects.

ACCESSIBLE DESCRIPTION

Three token cards connect to ID cards, which connect to highlighted rows in an embedding table.

Objective	Separate the token string, its numeric ID, and the embedding row the ID selects.
Recommendation	Lookup-key flow is clear and avoids implying the ID contains meaning.
Sample labels	token | ID | table row | dog | ID 421 | row | cat | ID 982 | row | floor | ID 1576 | row | ID is a lookup key, not meaning | 1 | 2 | 3

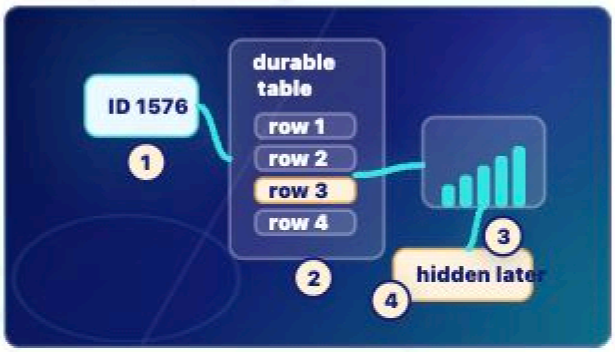
13 Embedding Lookup

embeddings · coded SVG · P3 · Keep

EMBEDDINGS

Embedding Lookup

Lesson: Embeddings | Pattern: vector | Variant: origami-object



Embedding Lookup

Durable table, temporary vector

A token ID retrieves a learned starting vector from a durable embedding table.

1. **Durable table** The embedding table was learned during training.
2. **Temporary retrieval** Inference retrieves one row for the current context.
3. **Starting vector** The retrieved vector starts the token in numerical space.
4. **Hidden state later** Transformer layers later reshape it into context-shaped hidden states.

> **Key takeaway:**
> Embedding means learned starting vector, not dictionary definition.

LEARNING OBJECTIVE

Distinguish the durable learned table from the temporary vector retrieved for the current run.

ACCESSIBLE DESCRIPTION

Objective	Distinguish the durable learned table from the temporary vector retrieved for the current run.
Recommendation	Durable table versus temporary vector is clear.
Sample labels	ID 1576 | durable | table | row 1 | row 2 | row 3 | row 4 | hidden later | 1 | 2 | 3 | 4

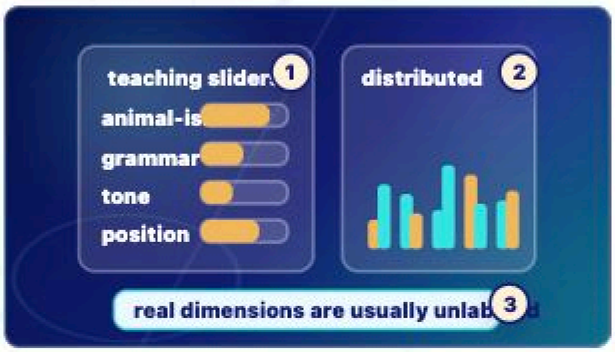
14 Feature Vector

vectors · coded SVG · P2 · Revise later

VECTORS

Feature Vector

Lesson: Vectors | Pattern: bars | Variant: origami-object



Feature Vector

Teaching labels versus distributed features

A vector is a list of numbers; real features are distributed rather than neat English sliders.

1. **Teaching sliders** Labels such as animal-ish or tone are simplified for learning.
2. **Distributed features** Real model features are spread across many dimensions.
3. **Why vectors** Vectors let the model compute with fuzzy numerical relationships.

> **Key takeaway:**
> Vectors are useful because many fuzzy features can vary at once.

LEARNING OBJECTIVE

Let learners compare a simplified slider view with unlabeled distributed numerical features.

ACCESSIBLE DESCRIPTION

The visual contrasts labeled teaching sliders with unlabeled distributed bars and dots.

Objective	Let learners compare a simplified slider view with unlabeled distributed numerical features.
Recommendation	Abstract feature dimensions are correct but could use a less dense future visual for nontechnical learners.
Sample labels	teaching sliders | animal-ish | grammar | tone | position | distributed | real dimensions are usually unlabeled | 1 | 2 | 3

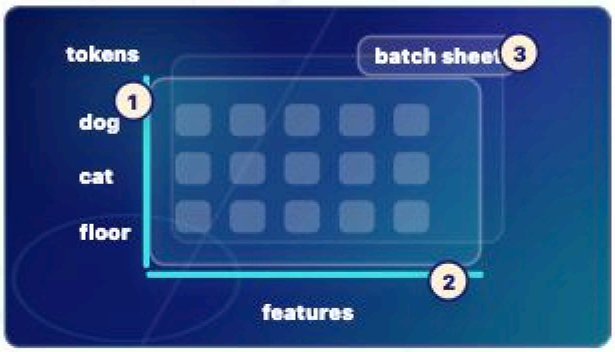
15 Tensor Block

tensors · coded SVG · P2 · Revise later

TENSORS

Tensor Block

Lesson: Tensors | Pattern: tensor | Variant: origami-object



Tensor Block

Token axis plus feature axis

Tensors organize many token vectors so model layers can process them.

1. **Token axis** Each row can represent one visible token position.
2. **Feature axis** Each column can represent one vector dimension.
3. **Batch note** Systems may process shapes like batch x tokens x features.

> **Key takeaway:**
> Tensors are shaped numerical blocks, not mystery objects.

LEARNING OBJECTIVE

Show token positions, feature dimensions, and an optional batch note without a heavy 3D library.

ACCESSIBLE DESCRIPTION

A lightweight axis diagram shows token labels down the side, feature labels across the top,

Objective	Show token positions, feature dimensions, and an optional batch note without a heavy 3D library.
Recommendation	Tensor blocks are accurate but visually abstract; future pass can clarify dimensions and temporary/durable status.
Sample labels	dog | cat | floor | tokens | features | batch sheet | 1 | 2 | 3

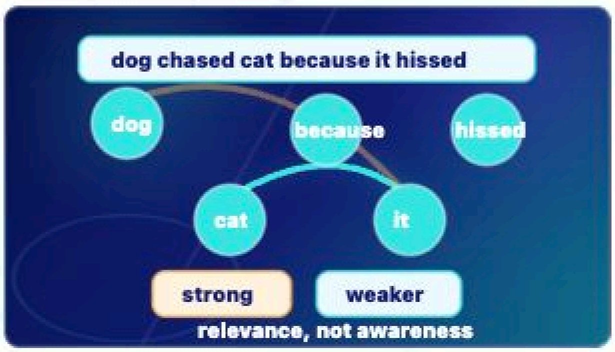
16 Attention Weave

attention · coded SVG · P3 · Keep

ATTENTION

Attention Weave

Lesson: Attention | Pattern: attention | Variant: paper-diagram



Attention Weave

Relevance, not awareness

Attention assigns temporary relevance weights between token positions in the current context.

1. **Current context** The visible tokens include dog, cat, because, it, and hissed.
2. **Strong relation** In this sentence, it is most relevant to cat.
3. **Weaker relation** dog is nearby and possible in other sentences, but it hissed points to cat here.
4. **Limit** Attention computes relevance weights. It is not human awareness.

> **Key takeaway:**
> Attention is relevance between token positions, not consciousness.

LEARNING OBJECTIVE

Use a concrete sentence to show that attention weights token relevance without implying awareness.

Objective	Use a concrete sentence to show that attention weights token relevance without implying awareness.
Recommendation	Attention arcs support weighted relevance without implying awareness.
Sample labels	dog chased cat because it hissed | dog | cat | because | it | hissed | strong | weaker | relevance, not awareness

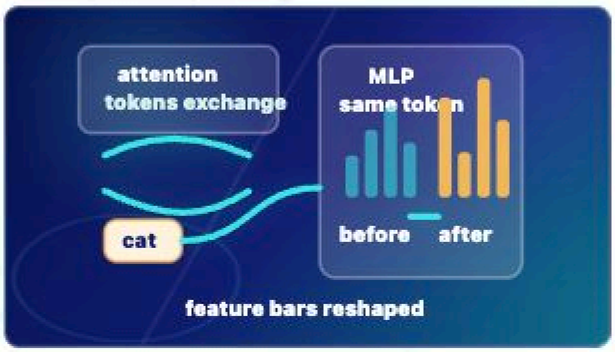
17 MLP Feature Workshop

mlp · coded SVG · P3 · Keep

MLP

MLP Feature Workshop

Lesson: MLP
Pattern: mlp
Variant: origami-object



MLP Feature Workshop

Same token, reshaped features

After attention mixes across tokens, the MLP reshapes features within each token position.

1. **Attention first** Token positions can exchange current-context information.
2. **Same position** The example token cat stays in the same position.
3. **MLP reshape** Feature bars change for that token position.
4. **Temporary values** The MLP has durable weights, but the feature values during inference are temporary activations.

> **Key takeaway:**
> Attention mixes across positions; the MLP reshapes features within a position.

LEARNING OBJECTIVE

Show the attention-versus-MLP distinction: attention mixes token positions; MLP reshapes each token vector independently.

Objective	Show the attention-versus-MLP distinction: attention mixes token positions; MLP reshapes each token vector independently.
Recommendation	Per-token feature reshaping is communicated with concise labels.
Sample labels	attention | tokens exchange | cat | MLP | same token | before | after | feature bars reshaped

18 Transformer Stack

layers · coded SVG · P3 · Keep

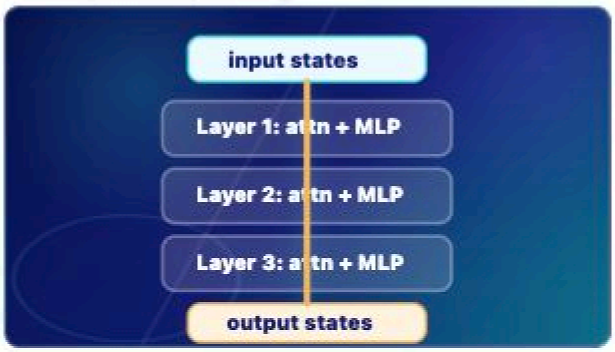
LAYERS

Transformer Stack

Lesson: Layers

Pattern: layers

Variant: origami-object



Transformer Stack

Repeated attention plus MLP

Layers repeat attention and MLP transformations to refine temporary hidden states.

1. **Input states** Current-context hidden states enter the stack.
2. **Repeated block** Each layer applies attention and an MLP, with residual paths and normalization helping signal flow.
3. **Output states** Refined hidden states continue toward next-token scores.
4. **Limit** Layers are repeated numerical transformations, not conscious reasoning steps.

> **Key takeaway:**
> Layers refine hidden states; they do not store permanent memory.

LEARNING OBJECTIVE

Show repeated transformer blocks without implying human thought steps.

Objective	Show repeated transformer blocks without implying human thought steps.
Recommendation	Stacked transformations remain readable and emphasize repetition.
Sample labels	input states | Layer 1: attn + MLP | Layer 2: attn + MLP | Layer 3: attn + MLP | output states

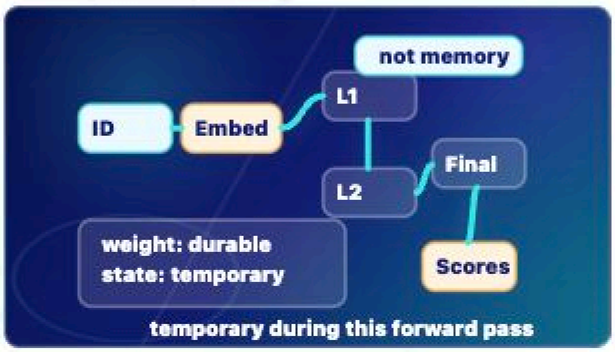
19 Hidden State Flow

hidden-states · coded SVG · P3 · Keep

HIDDEN-STATES

Hidden State Flow

Lesson: Hidden States | Pattern: hidden | Variant: origami-object



Hidden State Flow

Temporary during this forward pass

A token starts with an embedding and becomes context-shaped hidden states as it moves through layers.

1. **Embedding** A learned starting vector is retrieved for a token ID.
2. **Hidden states** Layers reshape that vector into temporary context-shaped states.
3. **Scores later** Final hidden states help produce next-token scores.
4. **Not memory** A hidden state is not visible text, permanent memory, or a stored fact.

> **Key takeaway:**
> Hidden states are temporary internal vectors shaped by current context.

LEARNING OBJECTIVE

Contrast embedding, hidden state, weight, memory, and visible text without making hidden states mysterious.

Objective	Contrast embedding, hidden state, weight, memory, and visible text without making hidden states mysterious.
Recommendation	Temporary context-shaped state is visually distinct from visible text.
Sample labels	ID | Embed | L1 | L2 | Final | Scores | weight: durable | state: temporary | not memory | temporary during this forward pass

20 Raw Scoreboard

logits · coded SVG · P3 · Keep

LOGITS

Raw Scoreboard

Lesson: Logits

Pattern: logits

Variant: paper-diagram

Raw Scoreboard

Before probabilities

The final hidden state feeds candidate next-token scores. These logits rank candidates, but they are raw scores, not probabilities or truth.

1. **Hidden state arrives** The final context-shaped vector reaches the output scoring step.
2. **Candidates get scores** Possible next tokens such as floor, room, counter, and quantum receive raw logits.
3. **Scores rank candidates** Higher score means stronger local candidate, not guaranteed truth.
4. **Softmax comes next** Softmax converts raw scores into probabilities.

> **Key takeaway:**
> Logits are raw next-token scores, not probabilities and not truth.

LEARNING OBJECTIVE

Objective	Show final hidden state becoming raw vocabulary scores before softmax.
Recommendation	Raw score bars clearly precede probabilities and truth checks.
Sample labels	hidden state | score | next? | floor | 8.2 high | room | 5.1 med | counter | 2.0 low | quantum | -1.0 v low | raw scores, not probabilities

21 Score to Probability

softmax · coded SVG · P3 · Keep

SOFTMAX

Score to Probability

Lesson: Softmax | Pattern: softmax | Variant: neon-flow

Score to Probability

Normalize before choosing

Softmax converts raw logits into probabilities that sum to 100%. High probability means likely under the model, not guaranteed true.

1. **Logits go in** Raw scores such as 8.2, 5.1, 2.0, and -1.0 enter softmax.
2. **Softmax converts** The function turns score differences into a probability distribution.
3. **Probabilities come out** The rounded teaching values sum to 100%.
4. **Not truth** Probability is about model likelihood, not evidence or correctness.

> **Key takeaway:**
> Softmax turns raw scores into probabilities; it does not verify truth.

LEARNING OBJECTIVE

Show raw-score values becoming rounded probabilities that sum to one.

Objective	Show raw-score values becoming rounded probabilities that sum to one.
Recommendation	Raw scores to probabilities is readable and mathematically bounded.
Sample labels	logits | probabilities | soft | max | floor | 8.2 | 86% | room | 5.1 | 12% | counter | 2.0 | 2% | quantum | -1.0 | ~0% | sum = 100% | probability != truth

22 Weighted Token Choice

sampling · coded SVG · P3 · Keep

SAMPLING

Weighted Token Choice

Lesson: Sampling | Pattern: sampling | Variant: neon-flow

Weighted Token Choice

One token chosen

Sampling chooses one next response token from probabilities. Probability is not truth: a low-probability token can still be possible, and a likely token can still be unsupported.

1. **Probability candidates** floor is strongest, room is plausible, counter is weak, and quantum is barely likely in this context.
2. **One token chosen** The decoding step chooses one next response token.
3. **Append next** The chosen token is appended before the model runs again.
4. **Not truth** A likely token can still be unsupported or wrong in a wider factual claim.

> **Key takeaway:**
> Sampling chooses one token from probabilities; probability is not truth.

Objective	Show probability weights and the single selected next token before autoregression repeats.
Recommendation	Weighted choice visual supports probability-not-truth distinction.
Sample labels	A jealous dog chased a startled cat | across the kitchen | floor | 86% | room | 12% | counter | 2% | quantum | ~0% | floor | chosen token | low probability is still possible

23 Append and Repeat

autoregression · coded SVG · P3 · Keep

AUTOREGRESSION

Append and Repeat

Lesson: Autoregression | Pattern: loop | Variant: neon-flow

Append and Repeat

One token, append, run again

The chosen token "floor" is appended to the response-so-far, then the model runs again.

1. **Response so far** The model has generated: A jealous dog chased a startled cat across the kitchen.
2. **Choose token** Sampling chooses one next token, here "floor".
3. **Append** The chosen token becomes part of the temporary current context.
4. **Run again** The next forward pass sees the longer response-so-far and chooses again.

> **Key takeaway:**
> Autoregression is choose, append, repeat; it is not permanent learning.

LEARNING OBJECTIVE

Make the autoregressive loop concrete: choose one token, append it, run the model again, and grow the response.

Objective	Make the autoregressive loop concrete: choose one token, append it, run the model again, and grow the response.
Recommendation	Loop visual clearly shows choose, append, repeat.
Sample labels	User prompt: pets in conflict | response so far | ...cat across the kitchen | 1 | floor | 2 | next context | prompt + response so far + floor | 3 | choose -> append -> run again

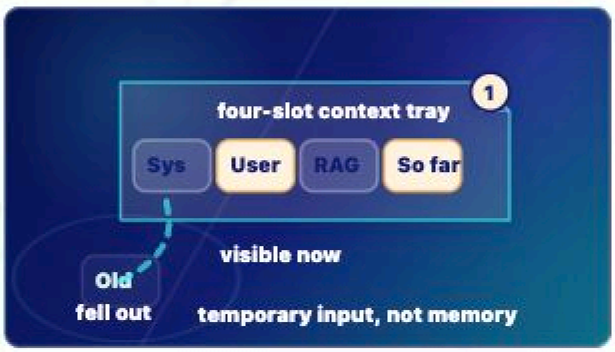
24 Temporary Context Window

context-window · coded SVG · P3 · Keep

CONTEXT-WINDOW

Temporary Context Window

Lesson: Context Window | Pattern: window | Variant: paper-diagram



Temporary Context Window

Limited visible input

Only the cards still inside the context tray can influence the next token.

1. **Visible now** System instructions, the user prompt, retrieved notes, and response-so-far may all be current context.
2. **Limited tray** When the tray fills, older material can fall out or need summarizing by the surrounding system.
3. **Temporary** Context can feel like memory, but it does not normally update weights or create permanent memory.
4. **Next token** The model can use only what remains visible in the current run.

> **Key takeaway:**
> Context is temporary visible input, not durable memory.

LEARNING OBJECTIVE

Objective	Show the context window as temporary capacity: system, user, retrieved, and response-so-far cards can fit, while old context can fall out.
Recommendation	Temporary context capacity and fall-out behavior are easy to read.
Sample labels	four-slot context tray | 1 | Sys | User | RAG | So far | Old | fell out | visible now | temporary input, not memory

25 Open-Book Retrieval

rag-retrieval · coded SVG · P3 · Keep

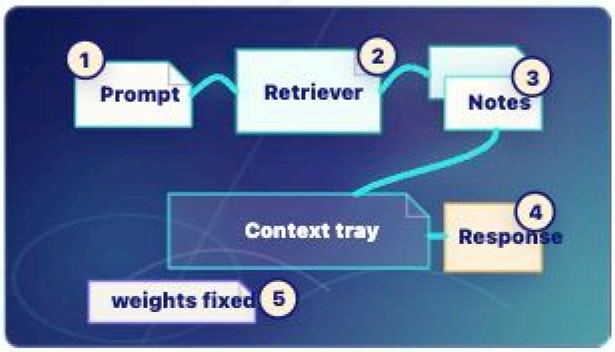
RAG-RETRIEVAL

Open-Book Retrieval

Lesson: RAG and Retrieval

Pattern: rag

Variant: retrieval-shelf



Open-Book Retrieval

Retrieval plus context, not training

Retrieved notes enter the context before response tokens are generated.

1. **Ask** The user prompt starts the run.
2. **Retrieve** A search system finds relevant outside material.
3. **Add to context** Retrieved notes become temporary context tokens.
4. **Generate** The model still generates response tokens one at a time.
5. **Weights stay fixed** RAG does not normally update model weights.

> **Key takeaway:**
> RAG is retrieval plus context, not training.

LEARNING OBJECTIVE

Show that RAG retrieves outside information and places it into context; it does not train the model.

Objective	Show that RAG retrieves outside information and places it into context; it does not train the model.
Recommendation	Open-book retrieval flow clearly places retrieved text into context.
Sample labels	Prompt | 1 | Retriever | 2 | Notes | 3 | Context tray | Response | 4 | weights fixed | 5

26 Claim Support Map

grounding-evidence · coded SVG · P3 · Keep

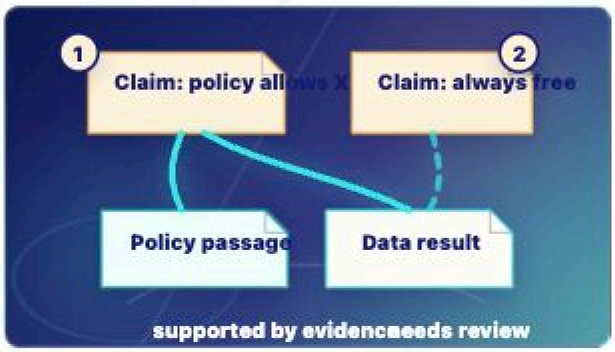
GROUNDING-EVIDENCE

Claim Support Map

Lesson: Grounding

Pattern: groundingEvidence

Variant: retrieval-shelf



Claim Support Map

Tying answers to evidence

Grounding asks whether generated claims are actually connected to available evidence.

1. **Claim** A generated answer can contain several claims.
2. **Evidence** Retrieved passages, data, citations, or tool results can support some claims.
3. **Support check** A citation-looking answer is not grounded unless the evidence actually matches the claim.
4. **Limit** Grounding helps, but the evidence and its use still need review.

> **Key takeaway:**
> Grounding ties claims to evidence; it does not make every claim true.

LEARNING OBJECTIVE

Show grounding as a claim-to-evidence support relationship, not a truth guarantee.

Objective	Show grounding as a claim-to-evidence support relationship, not a truth guarantee.
Recommendation	Claim-to-evidence structure reinforces grounding without a truth guarantee.
Sample labels	Claim: policy allows X | 1 | Claim: always free | 2 | Policy passage | Data result | supported by evidence | needs review

27 Unsupported Bridge

hallucination-bridge · coded SVG · P3 · Keep

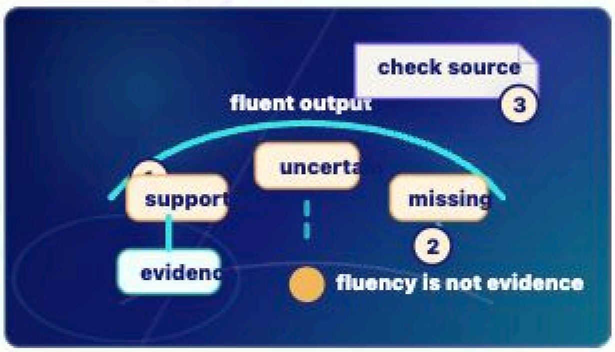
HALLUCINATION-BRIDGE

Unsupported Bridge

Lesson: Hallucinations

Pattern: hallucinationBridge

Variant: zen-garden-map



Unsupported Bridge

Fluent is not always grounded

A response can look smooth while one claim has support, another is uncertain, and another lacks evidence.

1. **Fluent surface** The output may read smoothly and confidently.
2. **Supported claim** One claim can be tied to evidence.
3. **Missing support** Another claim may lack evidence, citation, or retrieved context.
4. **Review** Grounding, uncertainty, and human review reduce risk but do not erase it.

> **Key takeaway:**
> Fluency is not evidence.

LEARNING OBJECTIVE

Show hallucination as fluent generated output without enough evidence support, without implying lying or intent.

Objective	Show hallucination as fluent generated output without enough evidence support, without implying lying or intent.
Recommendation	Unsupported bridge metaphor communicates fluent output outrunning support.
Sample labels	fluent output | 1 | supported | uncertain | missing | evidence | 2 | check source | 3 | fluency is not evidence

28 Learning Modes Matrix

ai-learns · coded SVG · P2 · Revise later

AI-LEARNS

Learning Modes Matrix

Lesson: How AI Learns

Pattern: learns

Variant: paper-diagram

Learning Modes Matrix

Durable change or current-run steering

Training, fine-tuning, feedback, prompting, RAG, and evaluation affect different parts of an AI system.

1. **Training** Pretraining and fine-tuning can durably update weights or adapter weights.
2. **Prompting** Prompts and in-context examples steer the current run.
3. **RAG** Retrieval adds outside material to context; it does not normally train the model.
4. **Evaluation** Review and tests can help future systems when results feed development or training.

> **Key takeaway:**
> Not all useful AI behavior is durable learning.

LEARNING OBJECTIVE

Compare ways systems change, retrieve, or are steered without implying every useful response updates model weights.

Objective	Compare ways systems change, retrieve, or are steered without implying every useful response updates model weights.
Recommendation	Matrix is accurate but dense; future pass should make durable versus temporary categories more scannable.
Sample labels	mode | weights? | context? | future? | when | Training | yes | no | yes | before | Fine-tune | yes | no | yes | after | Feedback | maybe | no | yes | review | Prompt | no | yes | no | run | RAG | no | yes | no | run

29 Append Or Denoise

diffusion · coded SVG · P3 · Keep

DIFFUSION

Append Or Denoise

Lesson: Diffusion vs Autoregression

Pattern: diffusion

Variant: zen-garden-map

Append Or Denoise

Two generation patterns

Autoregressive text generation appends tokens; diffusion-style generation refines noisy patterns.

1. **Autoregression** The text response grows by choosing and appending one token at a time.
2. **Diffusion** A noisy representation is refined step by step toward an output.
3. **Both generative** Both can create new outputs, but the generation patterns differ.

> **Key takeaway:**
> Generative AI is not one mechanism.

LEARNING OBJECTIVE

Make LLM token-by-token generation visibly different from diffusion denoising.

ACCESSIBLE DESCRIPTION

A split diagram shows an autoregressive lane with token, append, token, append and a

Objective	Make LLM token-by-token generation visibly different from diffusion denoising.
Recommendation	Append-versus-denoise contrast is direct and avoids ChatGPT-as-all-AI.
Sample labels	Autoregression | A jealous | dog | floor | append | Diffusion | noise | rough | clear | final | denoise and refine

30

Media Lane Map

multimodal · coded SVG · P3 · Keep

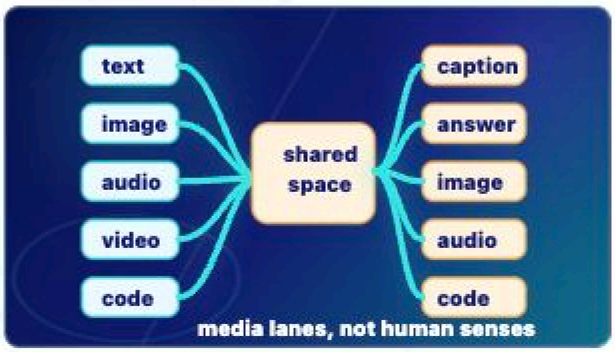
MULTIMODAL

Media Lane Map

Lesson: Multimodal AI

Pattern: multimodal

Variant: paper-diagram



Media Lane Map

Input, representation, output

Multimodal systems connect text, images, audio, video, and code through learned representations or linked components.

1. **Inputs** Text, image, audio, video, or code can enter the system.
2. **Representation** The system encodes media into model-usable representations or linked spaces.
3. **Outputs** Depending on the system, outputs may be answers, captions, images, audio, or code.
4. **Limit** Multimodal processing is not human sensation or experience.

> **Key takeaway:**
> Multimodal means more than one media type, not human-like perception.

LEARNING OBJECTIVE

Show multiple media inputs and outputs

Objective	Show multiple media inputs and outputs without implying human-like perception.
Recommendation	Media lanes are readable and avoid human-senses overclaim.
Sample labels	text | image | audio | video | code | shared | space | caption | answer | image | audio | code | media lanes, not human senses

31 Storm Front

perfect-storm · coded SVG · P3 · Keep

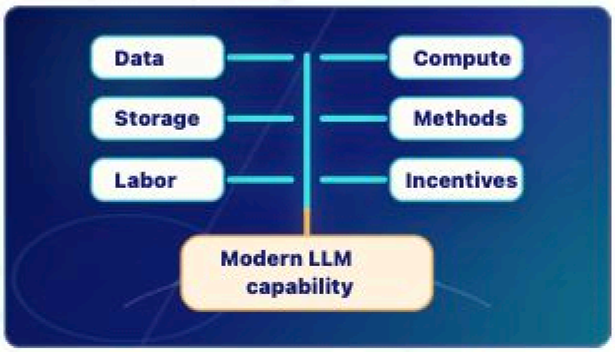
PERFECT-STORM

Storm Front

Lesson: The Perfect Storm

Pattern: perfectStorm

Variant: zen-garden-map



Convergence, not one spark.

Storm Front

Why LLMs arrived now

Data, compute, storage, methods, labor, and incentives converged into modern LLM capability.

1. **Data** Human-created text, media, code, and documents supplied patterns.
2. **Compute and storage** Hardware, data centers, datasets, and checkpoints made large-scale training practical.
3. **Methods** Deep learning and transformer advances made the patterns usable.
4. **Labor and incentives** Human evaluation work and market demand pushed systems into products.

> **Key takeaway:**
> Modern LLMs came from a convergence, not a single spark.

LEARNING OBJECTIVE

Show convergence without implying one magic

Objective	Show convergence without implying one magic breakthrough.
Recommendation	Repaired convergence layout is readable and keeps “convergence, not spark” out of the diagram body.
Sample labels	Data | Compute | Storage | Methods | Labor | Incentives | Modern LLM | capability

32 Borrowed Flames

collective-intelligence-lantern · coded SVG · P1 · Revise later

COLLECTIVE-INTELLIGENCE-LANTERN

Borrowed Flames

Lesson: Collective Intelligence, Extracted

Pattern: collectiveLantern

Variant: zen-garden-map

Borrowed Flames

No creators, no model

Human-created traces can become training data and model patterns, while rights and responsibility remain human questions.

1. **Human expression** Books, sites, code, art, journalism, forums, documentation, and research leave learnable traces.
2. **Collection questions** Provenance, consent, copyright, attribution, and compensation matter.
3. **Model limit** The model absorbs statistical patterns; it does not understand gratitude or responsibility.

Key takeaway:
The model did not create its abilities alone.

LEARNING OBJECTIVE

Make human-created source traces visible

Objective	Make human-created source traces visible without treating the model as humanity’s mind.
Recommendation	Important provenance/labor idea is still abstract; future visual should make data, labor, and extraction relationships more concrete.
Sample labels	Writing | Code | Art | Research | Forums | Docs | Training | data | rights questions

33 Benefit Tiers

benefits-tool-garden · coded SVG · P1 · Revise later

BENEFITS-TOOL-GARDEN

Benefit Tiers

Lesson: Benefits Worth Taking Seriously

Pattern: benefitsGarden

Variant: zen-garden-map

Benefit Tiers

Benefits without utopia

AI benefits are easiest to trust when claims are tiered by evidence, context, safeguards, and human review.

1. **Useful now with review** Accessibility, translation support, summarization, search/RAG, drafting, and coding assistance can help under human review.
2. **Plausible with safeguards** Tutoring support, research triage, brainstorming, and workflow support should be evaluated in context.
3. **Speculative / hype** Broad utopia, replacement, or no-cost claims should not be stated as fact.

> **Key takeaway:**
> Benefits can be real and still bounded.

LEARNING OBJECTIVE

Objective	Separate useful reviewed benefits, plausible guarded benefits, and speculative hype.
Recommendation	Benefit tiers need sharper connection to realistic use plus human review.
Sample labels	Benefit confidence tiers | Useful now | + review | Access | Translate | Summarize | Search/RAG | Plausible | + safeguards | Tutoring | Triage | Workflow | Speculative | / hype | Solves all | Replaces judgment | No costs | Real help needs evidence + review

34 Cost Ledger

costs-invisible-factory · coded SVG · P1 · Revise later

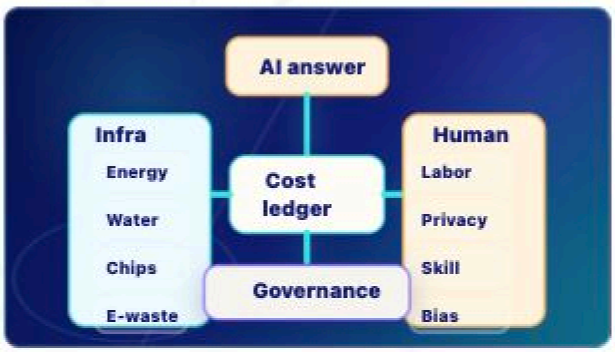
COSTS-INVISIBLE-FACTORY

Cost Ledger

Lesson: Costs We Must Count

Pattern: costsFactory

Variant: zen-garden-map



Cost Ledger

The answer is not weightless

AI systems depend on infrastructure, human systems, and governance choices that need honest accounting.

1. **Infrastructure** Energy, water, carbon, data centers, chips, and e-waste vary by system and workload.
2. **Human systems** Labor disruption, deskilling, privacy, bias, and information pollution depend on deployment choices.
3. **Power** Concentration of data, compute, and capital affects who benefits and who decides.

> **Key takeaway:**
> Costs vary, but they are real enough to count.

LEARNING OBJECTIVE

Make AI costs visible as a ledger without inventing statistics or fear-heavy imagery.

ACCESSIBLE DESCRIPTION

Objective	Make AI costs visible as a ledger without inventing statistics or fear-heavy imagery.
Recommendation	Cost categories are important and can become crowded; future pass should make costs easier to scan.
Sample labels	AI answer | Cost | ledger | Infra | Energy | Water | Chips | E-waste | Human | Labor | Privacy | Skill | Bias | Governance

35 Accountability Flow

human-centered-ai-garden · coded SVG · P1 · Revise later

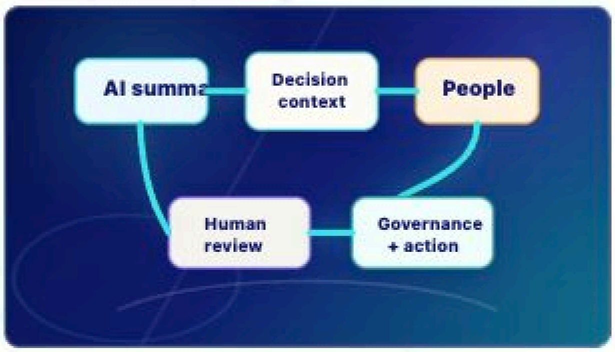
HUMAN-CENTERED-AI-GARDEN

Accountability Flow

Lesson: Human-Centered AI

Pattern: humanGarden

Variant: zen-garden-map



Accountability Flow

Tools should serve dignity

A human-centered deployment treats AI output as support inside a decision context, then keeps review, governance, and accountable action with people.

1. **Human judgment** People remain accountable for high-stakes decisions and institutional use.
2. **Dignity** Speed, profit, and automation should not outrank persons, learning, or relationships.
3. **Model limit** A model can sound ethical without moral understanding.

> **Key takeaway:**
> Powerful tools still need human purpose.

LEARNING OBJECTIVE

Make human-centered AI concrete through a support-note accountability scenario.

ACCESSIBLE DESCRIPTION

Objective	Make human-centered AI concrete through a support-note accountability scenario.
Recommendation	Accountability flow should be made more operational and less metaphor-dependent.
Sample labels	AI summary | Decision | context | People | Human | review | Governance | + action

36 Better-AI Control Panel

responsible-ai-forked-path · coded SVG · P1 · Revise later

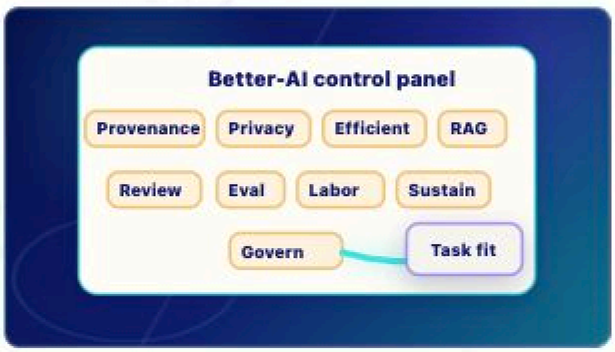
RESPONSIBLE-AI-FORKED-PATH

Better-AI Control Panel

Lesson: Better AI Is a Choice

Pattern: responsiblePath

Variant: zen-garden-map



Better-AI Control Panel

Responsible AI is chosen

AI outcomes are shaped by design, deployment, governance, incentives, and institutional choices.

1. **Technical choices** Smaller models, efficient inference, distillation, RAG, and better hardware use can fit some tasks.
2. **Data choices** Provenance, consent, licensing, creator compensation, and privacy-preserving deployment matter.
3. **Institutional choices** Human review, policy, labor transition planning, public-interest models, and independent evaluation shape outcomes.

> **Key takeaway:**
> Harms are shaped by choices, not destiny.

LEARNING OBJECTIVE

Show practical design and governance levers

Objective	Show practical design and governance levers without pretending governance is simple.
Recommendation	Governance/control-panel idea should be clearer for institutional decision makers.
Sample labels	Better-AI control panel | Provenance | Privacy | Efficient | RAG | Review | Eval | Labor | Sustain | Govern | Task fit

37 Context Tray

prompting-context-tray · coded SVG · P3 · Keep

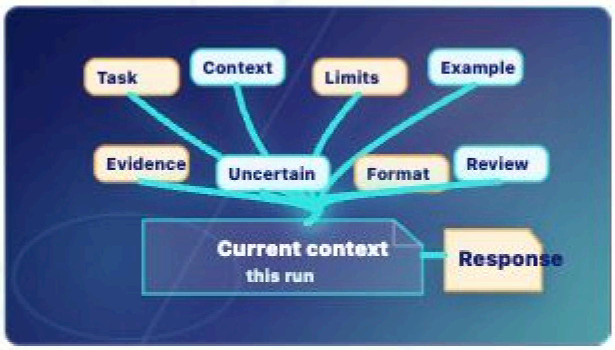
PROMPTING-CONTEXT-TRAY

Context Tray

Lesson: Effective Prompting from Model Literacy

Pattern: promptingTray

Variant: retrieval-shelf



Context Tray

Prompting steers this run

Good prompts pack task, context, constraints, examples, evidence needs, uncertainty, review, and format into the current run.

1. **Prompt parts** Task, context, constraints, examples, evidence needs, uncertainty, review, and format shape the current input.
2. **Evidence and uncertainty** Source needs, retrieved context, and uncertainty requests can improve reviewability.
3. **Boundary** Prompting usually changes context, not weights.

> **Key takeaway:**
> Prompting is context design for one run.

LEARNING OBJECTIVE

Show prompting as context design, not permanent teaching.

ACCESSIBLE DESCRIPTION

Objective	Show prompting as context design, not permanent teaching.
Recommendation	Context tray supports prompting-as-context-shaping.
Sample labels	Task | Context | Limits | Example | Evidence | Uncertain | Format | Review | Current context | this run | Response

38 Full Chain Map

synthesis-map-compass-lantern · coded SVG · P1 · Revise later

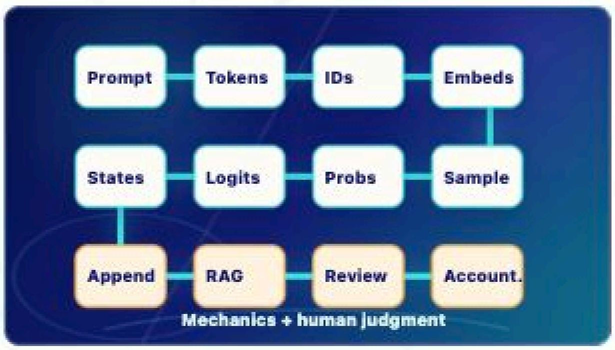
SYNTHESIS-MAP-COMPASS-LANTERN

Full Chain Map

Lesson: Model Literate Synthesis

Pattern: synthesisMap

Variant: zen-garden-map



Full Chain Map

Mechanics plus judgment

Model literacy connects prompt, tokens, IDs, embeddings, hidden states, logits, probabilities, sampling, append-and-repeat generation, RAG, grounding, review, and accountability.

1. **Machine chain** Prompt context becomes tokens and IDs, then embeddings, hidden states, logits, probabilities, and sampled response tokens.
2. **Evidence chain** Context windows, RAG, and grounding can add evidence for this run, but they do not guarantee truth.
3. **Human chain** Benefits, costs, review, governance, and accountability remain human responsibilities.

> **Key takeaway:**
> Mechanics matter, and humans remain responsible.

LEARNING OBJECTIVE

Objective	Close the Journey by connecting the machine chain with human consequences.
Recommendation	Full-chain synthesis is the densest visual; needs simplification before badge-facing review.
Sample labels	Prompt | Tokens | IDs | Embeds | States | Logits | Probs | Sample | Append | RAG | Review | Account. | Mechanics + human judgment

39 Risk Ledger

risk · coded SVG · P2 · Revise later

RISK

Risk Ledger

Lesson: Risk vs Myth

Pattern: risk

Variant: zen-garden-map

Risk Ledger

Mechanisms, not magic

Risk literacy separates practical, mechanism-based risks from myths about what models are.

1. **Real mechanisms** Privacy exposure, hallucinations, prompt injection, insecure tools, overreliance, and bias can cause practical harm.
2. **Myths** Conscious chatbots, omniscient databases, secret self-training during every chat, and softmax stealing files are magical stories.
3. **Human accountability** Institutions, vendors, policies, and affected communities shape how risks are prevented and reviewed.

> **Key takeaway:**
> Real AI risks usually come from data, context, tools, institutions, and overreliance, not model consciousness or magic.

Objective	Separate concrete institutional risks from magical stories without doom or hype.
Recommendation	Risk ledger is useful but can be refined so real risks and myths remain balanced and readable.
Sample labels	Real risk | Myth / magic | Privacy | Halluc. | Prompt inj. | Tool risk | Overrely | Bias | Conscious | Self-trains | Omniscient | Softmax files